

TECHNICAL SPECIFICATION FOR 11 KV COMBINED CURRENT & POTENTIAL TRANSFORMER SUITABLE FOR THREE ELEMENT METER

1. SCOPE :

This specification covers the design, manufacture, assembly, testing at the manufacturer's works, supply & delivery at WBSEDCL's stores/sites anywhere in WB including unloading of combined current & potential transformers of 11KV voltage class as specified in Schedule - A for metering services in three phase system for three element meter.

2. STANDARD :

The combined current & potential transformers unit and accessories covered by this specification shall comply with the requirement of the latest edition of the following standards unless otherwise stated in this specification.

IS:16227 (Part-1 to 4) : Specification for Instrument Transformers.

IS:2099 & IS:5621 : Specification for HV Porcelain bushings.

3. DEVIATION :

Normally the offer should be as per Technical Specification without any deviation. But any deviation felt necessary to improve performance, efficiency and utility of equipment must be mentioned in the Deviation Schedule with reasons duly supported by documentary evidence. Such deviations suggested may or may not be accepted by the Company. Deviations not mentioned in Deviation schedule will not be considered.

4. DESIGN & CONSTRUCTION OF 11 KV COMBINED CURRENT & POTENTIAL TRANSFORMERS:

The design features and construction details of combined current & potential transformer shall be in accordance with the requirement stipulated in Schedule - A :

- i) The combined current & potential transformers shall be complete in all respects and shall conform to the modern practice of design and manufacture.
- ii) The combined current & potential transformers shall be of low Reactance outdoor type, three phase, 50 Hz, self cooled with shaded porcelain bushing suitable for operation under the service conditions as specified in the general condition of site suitable for outdoor operation.
- iii) The maximum permissible temperature rise of the combined current & potential transformer winding when carrying a primary current equal to the rated continuous current at rated frequency and with rated burden over an ambient temperature shall not exceed 55°C.
- iv) The combined current and potential transformer shall be oil cooled type .The combined current & potential transformers shall be suitable for up right mounting on PCC POLE DP structures.
- v) The combined CT & PT Unit shall be complete with all accessories like primary terminal connectors, weather proof terminal box for secondary connection, lifting lugs, grounding terminals and name plate.
- vi) The combined CT & PT Unit shall be oil cooled type provided with class - A insulation. It shall be of hermetically sealed type construction to prevent air & moisture from entering the tank. The design and construction of combined CT & PT Unit shall be sufficient to withstand the thermal and mechanical stresses resulting from the specified short circuit currents and specified duration as mentioned in General Technical Specification.
- vii) The core of the combined CT & PT Unit shall be high grade non-ageing, electrical, silicon laminated steel of low hysteresis loss and high permeability to ensure high accuracy at both normal and over current.
- viii) The exciting current shall be as low as possible and the combined CT & PT Unit shall be capable of maintaining its rated accuracy at different burdens and within saturation limits.
- ix) Rating plate marking shall be provided as per relevant clause of IS: 16227.
- x) The combined CT & PT Unit Characteristics shall be such as to provide satisfactory performance for burdens ranging from at least 25% to 100% of rated burden in case of metering combined CT & PT Unit cores.

- xi) When operating the current transformer between 5% of the rated current and the rated continuous thermal current, the voltage transformer shall not exceed the limits of voltage error and phase displacement corresponding to its class within the specified range of burden and between 80% and 120% of the rated voltage.
When operating the voltage transformer between 80% of the rated voltage and the rated voltage multiplied by the rated voltage factor, the current transformer shall not exceed the limits of current error and phase displacement within the range of current corresponding to its class and within the specified range of burden.
- xii) The combined CT & PT Unit secondary terminals shall be brought out in a weather proof terminal box. The terminal box shall be provided with glands suitable for 4 core 4 Sq. mm & 8 core 4 sq. mm 1.1 KV grade, steel armoured PVC sheathed stranded copper conductor cables.
- xiii) The combined CT & PT Unit secondary to be used for metering and instruments shall be of accuracy class and I S F as specified. The saturation factor of this core shall be low enough so as not cause any damage to measuring instruments in the event of maximum short circuit current.
- xiv) The ratio changing arrangement shall be provided on secondary side of the combined CT & PT Unit.

5. BUSHING :

- i) The insulation of bushings shall be co-ordinated with that of the combined CT & PT Unit such that the flashover, if any will occur only external to the combined CT & PT Unit.
- ii) Each of the bushings porcelain shall have creepage distance suitable for voltage class as specified in specified IS.

6. GROUNDING TERMINALS :

Two grounding terminals on diagonally opposite sides of adequate size suitable for connecting M.S.Flat of size 50 mm x 6 mm shall be provided.

7. SECONDARY TERMINAL BOX :

- i) All secondary terminals shall be brought out in a compartment on one side of the combined CT & PT Unit for easy access.
- ii) The exterior of this terminal box shall be painted.
- iii) Arrangement for shorting of CT secondary terminals shall be provided in the CT secondary terminal box with supply of shorting link made of copper.
- iv) The terminal box shall be provided with removable cable gland plate with provision of punching at bottom for mounting required number of cable glands of 1.1 KV grade steel wire armoured, PVC insulated PVC sheathed 4 sq.mm, 4core and 8 Core stranded copper conductor cables. The cable glands shall be included within the scope of supply.
- v) The terminal box shall be provided with flap type cover plate so as to have easy access of secondary terminals. The cover shall have a sealing/ locking arrangement and shall be suitable to prevent ingress of moisture and rain water.
- vi) All terminals shall be clearly marked with identification number to facilitate connection to external wiring with sufficient space in between.

8. TERMINAL CONNECTOR:

Aluminium Terminal Connector suitable for ACSR Dog Conductor is to be supplied along with Combined CT & PT unit.

9. PAINTING :

- i) The tank and top metallic cover shall be painted with epoxy paint. All steel surfaces shall be cleaned by sand blasting or chemical process as required to produce a smooth surface, free of scale, grease and dirt.
- ii) Paints shall be carefully selected to withstand tropical heat, rain etc. The paints shall not scale off or crinkle or be removed by abrasion due to normal handling.

10. TESTS AT FACTORY AND TEST CERTIFICATES :

- i) Each combined C.T/PT shall comply with the requirements of routine test as specified in the relevant Clause of IS:16227.
- ii) Routine test at manufacturer's works shall be carried out in presence of representative of WBSEDCL as per relevant IS.
- iii) All Acceptance tests shall be carried out at the manufacturer's works on every lot offered for inspection as per relevant IS. In addition to above, all routine tests are also to be carried out on combined CT & PT Unit as per relevant IS. Selection of samples for acceptance test as well as rejection and retesting shall be guided by relevant IS.

11. TEST REPORTS AND TYPE TESTS :

The Bidder should submit all the type test report on 11KV Combined CT & PT with CTR:-200-100/1A at higher ratio as per IS 16227 from CPRI/NABL accredited/Govt. recognized Test House or Laboratory carried out within 5 years from the due date of submission of bid having identical rating and type as that of the tendered item as pre-requisites along with the bid failing which their offer may not be technically acceptable. The Type Test certificates of the NABL accredited/govt. recognized test house or laboratory should however, bear the logo of NABL accreditation.

Each Type Test Report shall comply the following information with test results.

- i) Complete identification, date and serial no.
- ii) Method of application where applied, duration and interpretation of each test.

12. TENDER DRAWING, CATALOGUE AND TEST REPORTS :

One copy of the following drawings and catalogue shall be submitted with each copy of tender for evaluation :

- i) General arrangement drawings showing front elevation, side view, plan along with all accessories, mounting arrangement, creepage distance of the bushing, electrical diagram of primary and secondary connection with polarity marking, terminal arrangement of secondary terminal box, size of primary terminals, grounding terminals and lifting lugs, net and shipping weight, dimension etc.
- ii) Name and rating plate diagram.

13. CONTRACT DRAWINGS AND MANUALS:

13.1 In the event of placement of order the following drawings and manuals shall be submitted in six (6) copies to the Chief Engineer, P&C, WBSEDCL, Vidyut Bhavan (4th floor), Kolkata – 700091 for approval.

- a) General outline dimension drawing of current transformers furnishing front and side elevation, top and bottom plan, views showing all accessories, mounting arrangement on DP structures, including dimension of the bolts, total creepage distance of bushing, electrical diagram for primary and secondary connections with polarity mark, terminal arrangement for secondary terminal box, size of primary terminals, grounding terminals and lifting lugs, quantity of insulating oil, net and shipping weight, shipping dimension etc.
- b) Name and rating plate diagram of combined CT & PT Unit.

13.2 After approval, ten (10) sets of approved drawings and operating and maintenance manual including the instruction manual shall be submitted for our record and distribution to site. (Two complete sets should be directly sent to Chief Engineer, (Dist.), Vidyut Bhavan (1st floor), Kolkata – 700091)

Instruction manual should contain :

- a) A brief description of combined CT & PT Unit furnishing the constructional features.
- b) Instruction for handling, storing, erection, commissioning and operation and maintenance of combined CT & PT Unit
- c) General outline drawing of the combined CT & PT Unit along with all components and accessories.
- d) Marked erection points identifying the component parts of combined CT & PT Unit.
- e) Detailed dimensions of assembly and description of all accessories.
- f) Detailed views of Core, winding assembly, winding connections and its tappings.
- g) List of spares and other necessary information for combined CT & PT Unit
- h) A set of approved test certificate.
- i) All notes and legends of the drawings shall be furnished in English and all dimensions shall be marked in metric units.

14.00 Documents to be submitted at the time of physical delivery at consignee stores :

The following documents to be submitted by the Vendors to the Consignee Stores at the time of physical delivery :-

- a) Copy of Purchase Order.
- b) Copy of Despatch Instruction.
- c) Inspection Test Certificate.
- d) Guarantee Certificate.
- e) Proforma Invoice.
- f) Calculation Sheet for price Variation on the basis of IEEMA as applicable with base date of order.
- g) Seal list and packing list.
- h) Challan in triplicate.
- i) Way bill, if applicable.

SCHEDULE-A
SPECIFIC TECHNICAL PARTICULARS OF CURRENT TRANSFORMER

Sl. No.	Description	Rating
i)	Rated system voltage KV (rms)	11
ii)	Highest system voltage KV (rms)	12
iii)	CT Ratio	200-100/1A suitable for 3 Phase 4 Wire meter, rating -/1A & -/110V, 0.5 class(Three numbers CT is required)
iv)	Burden (VA)	10 VA
v)	Accuracy Class	0.5
vi)	System frequency (Hz)	50
vii)	System neutral earthing	Effectively Earthed
viii)	Installation	outdoor
ix)	Extended current rating	120 %
x)	Rated short time thermal current for 1 second (KA _{rms})	18.4
xi)	Rated Dynamic current KA (peak) (should be at least 2.5 times of above rating)	46
xii)	Rated Insulation level :	
	a) 1.2/50 microsecond impulse withstand voltage (KV _{peak})	75
	b) One minute power frequency withstand voltage (KV _{rms}) on primary winding	28
	c) Power frequency withstand voltage for secondary winding for one minute (KV _{rms})	3
xiii)	Over voltage interturn test	As per IS: 16227
xiv)	Minimum Creepage Distance (Heavily polluted atmosphere)(in mm)	300
xv)	Limit of temp. rise(°C) of windings at rated current	55°C
xvi)	Instrument security factor	Less than 5 for metering core at Lower Ratio
xvii)	Type of Insulation	Class A
xviii)	Mounting Dimension (mm) of CT and PT combined unit	243 X 243
xix)	Material and minimum Diameter of Primary stud of CT and PT combined unit	M-16, tin plated electrolytic copper
xx)	Painting shed of CT and PT combined unit	631 of IS-5
xxi)	Maximum Partial Discharge:	
	a) when energized at 12KV rms	10 PC
	b) when energized at 1.2x12/√3 KV rms	5 PC

SPECIFIC TECHNICAL PARTICULARS OF POTENTIAL TRANSFORMER

<u>Sl. No.</u>	<u>Description</u>	<u>Rating</u>
i)	Rated system voltage KV (rms)	11
ii)	Highest system voltage KV (rms)	12
Iii)	PT Ratio	11000/110 V
iv)	Burden (VA)	20 VA/phase
v)	Accuracy Class	0.5
vi)	System frequency (Hz)	50
vii)	Installation	Outdoor
viii)	No. of phase/ Connection	3 Ph, Star/star with both Neutral Earthed (3 phase 5 Limb PT)
ix)	Rated Insulation level:	
	a) 1.2/50 microsecond impulse withstand voltage (KV peak)	75
	b) One minute power frequency withstand voltage (KV rms) on primary winding	28
	c) Power frequency voltage withstand requirements for secondary winding for one minute (KV rms)	3
x)	Limit of temp. rise of winding (°C)	55° C above ambient temperature
xi)	Voltage factor	1.2 cont. & 1.9 for 8 hours
xii)	Maximum Partial Discharge:	
	a) when energized at 12KV rms	10 PC
	b) when energized at $1.2 \times 12 / \sqrt{3}$ KV rms	5 PC

GUARANTEED TECHNICAL PARTICULARS FOR CURRENT TRANSFORMER

<u>Sl No.</u>	<u>Description</u>	<u>Particulars</u>
1.	Make	
2.	Type	
3.	Reference Standard	
4.	Voltage Grade	
5.	Ratio	
6.	Frequency	
7.	No of core	
8.	Rated VA Burden	
9.	Accuracy Class	
10.	Class of Insulation	
11.	Temperature rise above ambient	
12.	Insulation level-KV (peak/rms)	
13.	a) Short time current rating(KA) for 1.0 sec b) Dynamic current rating(KA peak)	
14.	Instrument security factor	
15.	Magnetizing curve furnished	
16.	Mounting Dimension of combined CT and PT unit	
17.	Weight of Combined CT&PT Unit	
18.	Material and Diameter of the Primary Stud of the combined CT & PT unit	
19.	Degree of Protection of enclosure of combined CT and PT unit	
20.	Painting shed of Combined CT&PT Unit	
21.	Maximum Partial Discharge: a) when energized at 12KV rms b) when energized at $1.2 \times 12 / \sqrt{3}$ KV rms	

GUARANTEED TECHNICAL PARTICULARS FOR POTENTIAL TRANSFORMER

<u>Sl No.</u>	<u>Description</u>	<u>Particulars</u>
1.	Make	
2.	Type	
.	Reference Standard	
4.	Frequency	
5.	Rated Primary Voltage	
6.	Rated Secondary Voltage	
7.	Winding Connection	
8.	Rated VA Burden per phase	
9.	Accuracy Class	
10.	Class of Insulation	
11.	Temperature Rise above Ambient	
12.	Over Voltage Factor	
	a) Continuous	
	b) 8 Hours	
13.	Maximum Partial Discharge	
	a) when energized at 12KV rms	
	b) when energized at $1.2 \times 12 / \sqrt{3}$ KV rms	